DATA SET DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+0.1	+0.7	+0.3	-0.1	+0.5	+0.2	+0.5
Rank	1,2	7	4	1,2	5,6	3	5,6

a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.

b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

H_0 : 2个总体概率分布相同 $(M_d = 0)$

H_a : 〃 不同 $(M_d \neq 0)$

(a) $T^+ = 1.5 + 3 + 4 + 5.5 + 5.5 + 7 = 26.5$

$T^- = 1.5$

$n = 7$, $\frac{7 \times 8}{2} = 28$, $26.5 + 1.5 = 28$

$T = \min(T^+, T^-) = 1.5$

$n = 7$, $\alpha = 0.05$, $T_0 = 2$

$T = 1.5 \leq 2$

$\rightarrow \in RR$

$\rightarrow$ Reject H_0

(b) 结论是一致的.

若数据差值满足正态分布，配对t检验会比 Wilcoxon 检验更具统計功效；但若数据不满足正态分布 or sample 太小无法验证正态性，Wilcoxon 则更 safe.

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

DATA SET
DS1511

$n=9 \rightarrow n=8$

Rank	Month	A	B	$d=A-B$
6	1	3	7	−4
3.4	2	14	12	+2
3.4	3	7	9	−2
7	4	10	15	−5
5	5	9	12	−3
	6	6	6	0
1.2	7	13	12	+1
1.2	8	6	5	+1
8	9	7	13	−6

a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.

b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

H_0 : 2台 machine の月故障率分布没有显著差异 $(Md=0)$

H_a : 〃 存在显著差异 $(Md \neq 0)$

1a) $T^+ = 1.5 + 1.5 + 3.5 = 6.5$

$T^- = 3.5 + 5.0 + 6.0 + 7.0 + 8.0 = 29.5$

$T^+ + T^- = 6.5 + 29.5 = 36$, $\frac{n(n+1)}{2} = \frac{8 \times 9}{2} = 36$

$T = \min(T^+, T^-) = 6.5$

$n=8$, $\alpha = 0.05$, $T_0 = 4$

$T = 6.5 > 4$

$\rightarrow \notin RR$

$\rightarrow$ fail to reject H_0

1b) i Production Volume

ii Environmental Conditions

iii Maintenance Schedules

iv Operator Variance

DATA SET DS1514

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Rank	Student	With Imagery	Without Imagery		R	Student	With Imagery	Without Imagery	
16.5	1	20	5	15	5.5	11	17	8	9
16.5	2	24	9	15	1.0	12	20	16	4
16.5	3	20	5	15	7.5	13	20	10	10
5.5	4	18	9	9	2.0	14	16	12	4
19	5	22	6	16	20	15	24	7	17
3.5	6	19	11	8	13.5	16	22	9	13
11.5	7	20	8	12	3	17	25	21	4
3.5	8	19	11	8	1	18	21	14	7
7.5	9	17	7	10	2	19	19	12	7
11.5	10	21	9	12	11.5	20	23	13	10

a. What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?

b. Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.

c. Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

a. 3 testing : (1) t-test
(2) Wilcoxon signed-rank test
(3) sign test

Assumption : (1) independence
(2) 正态性

→ 不完全满足

b. Sign Test 只看 + - , 20个 student 差值全为 (+)

$\left(\frac{1}{2}\right)^{20} \approx 0.00000095$

p-value $= 2 \times 0.00000095 \approx 0.0000019$

p-value < 0.05 ⟶ Reject H_0

Wilcoxon $n = 20$

$T^+ = 1 + 2 + \cdots + 20 = \frac{20 \times 21}{2} = 210$

$T^- = 0$

$T = \min(T^+, T^-) = 0$

$n = 20$, $\alpha = 0.05$, $T_0 = 52$

$T = 0 \leq 52$ ⟶ $\in RR$

⟶ Reject H_0.

(c) The conclusions are same.

In 20 sample, 100% of student 在'有联想'情况下得分都高于'无联想'

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

DATA SET DS1518

Training Period (hours)			
.5	1.0	1.5	2.0
8	9	4	4
14	7	6	7
9	5	7	5
12	8	8	5
10	9	6	4

Use the Kruskal–Wallis H-Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

Time	Training Period	Rank
4	1.5	2
4	2	2
4	2	2
5	1	5
5	2	5
5	2	5
6	1.5	8
6	1.5	8
6	2.0	8
7	1	11
7	1.5	11
7	2.0	11
8	0.5	14
8	1	14
8	1.5	14
9	0.5	17
9	1.0	17
9	1.0	17
10	0.5	19
12	0.5	20
14	0.5	21

H_0 : four different lengths of instructional time 概率分布相同.

H_a : 至少一组 n 组装时间概率与其他组不同.

Group 1 : 0.5 h

$R_1 = 14+21+17+20+19 = 91$

Group 2 = 1 h

$R_2 = 17+11+5+14+17 = 64$

Group 3 : 1.5 h

$R_3 = 2+8+11+14+8 = 43$

Group 4 = 2 h

$R_4 = 2+11+5+5+2 = 25$

Total $R = 223$, $\frac{n(n+1)}{2} = \frac{20\times21}{2} = 210$

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(n+1)$$

$$= \frac{12}{20\times21}\left(\frac{91^2}{5} + \frac{64^2}{5} + \frac{43^2}{5} + \frac{25^2}{5}\right) - 3(21)$$

$= 21.86$

$H \approx 11.15$

$df = k-1 = 4-1 = 3$, $\alpha = 0.01$

$\chi^2_{0.01,3} = 11.345$

$\therefore 11.15 < 11.345$, $\notin RR$

$\longrightarrow$ Fail to reject H_0.

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

DATA SET DS1520

	10–19	20–39	40–59	60–69
	29	24	37	28
	33	27	25	29
	26	33	22	34
	27	31	33	36
	39	21	28	21
	35	28	26	20
	33	24	30	25
	29	34	34	24
	36	21	27	33
	22	32	33	32
Total	309	275	295	282

a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal–Wallis H-test with $\alpha = .01$.

b. Find the approximate p-value for the test in part a.

c. Since the F-test in Chapter 11 and the H-test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F-test produced the test statistic $F = 0.87$ with p-value $= 0.468$, compare this p-value with the p-value in part b and explain the implications of the comparison.

20	1	1	1.0
21	3	2, 3, 4	3.0
22	2	5, 6	5.5
24	3	7, 8, 9	8.0
25	2	10, 11	10.5
26	2	12, 13	12.5
27	3	14, 15, 16	15.0
28	3	17, 18, 19	18.0
29	3	20, 21, 22	21.0
30	1	23	23.0
31	1	24	24.0
32	2	25, 26	25.5
33	4	27, 28, 29, 30	28.5
34	2	31, 32	31.5
35	1	33	33.0
36	2	34, 35	34.5
37	1	36	36.0
39	1	37	37.0

$n = 40$, $\frac{40 \times 41}{2} = 820$

a. Group 1 (10-19)

$R_1 = 21 + 28.5 + 12.5 + 15 + 37 + 33 + 28.5 + 21 + 34.5 + 5.5 = 236.5$

Group 2 (20-39)

$R_2 = 8 + 15 + 28.5 + 24 + 3 + 18 + 8 + 31.5 + 3 + 25.5 = 164.5$

Group 3 (40-59)

$R_3 = 36 + 10.5 + 5.5 + 28.5 + 18 + 12.5 + 23 + 31.5 + 15 + 28.5 = 209$

Group 4 (60-69)

$R_4 = 18 + 21 + 31.5 + 34.5 + 3 + 1 + 10.5 + 8 + 28.5 + 25.5 = 210$

$\longrightarrow 236.5 + 164.5 + 209 + 210 = 820$

$$H = \frac{12}{n(n+1) \cdot 10} \Sigma R_i^2 - 3(n+1)$$

$$= \frac{12}{40 \times 41 \times 10} (236.5^2 + 164.5^2 + 209^2 + 210^2) - 3(41)$$

$$= 1.956$$

$df = k-1 = 3$, $\alpha = 0.01$, $\chi^2_{0.01,3} = 11.345$

$H = 1.956 < 11.345 \longrightarrow$ Fail to reject H_0

b. $\chi^2_{0.10,3} = 6.251$

$\chi^2_{0.50,3} = 2.366$

$\chi^2_{0.70,3} = 1.424$

$1.424 < 1.956 < 2.366$, p-value 介于 0.5 到 0.7

c. ANOVA $F = 0.87$, p-value $= 0.468$

Kruskal-Wallis $H = 1.956$, p-value $= 0.582$

$\longrightarrow$ 结論高度一致，都 fail to reject H_0.

10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.[7] The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

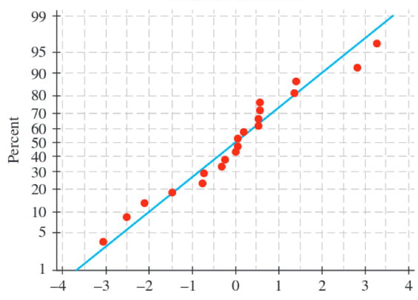
DATA SET DS1528

	Antibiotic			
Child	1	2	3	4
1	4.8 (2)	2.2 (1)	6.8 (4)	6.2 (3)
2	8.1 (2)	9.2 (3)	6.6 (1)	9.6 (4)
3	5.0 (3)	2.6 (1)	3.6 (2)	6.5 (4)
4	7.9 (2)	9.4 (4)	5.3 (1)	8.5 (3)
5	3.9 (3)	7.4 (4)	2.1 (2)	2.0 (1)
	$R_1=12$	$R_2=13$	$R_3=10$	$R_4=15$

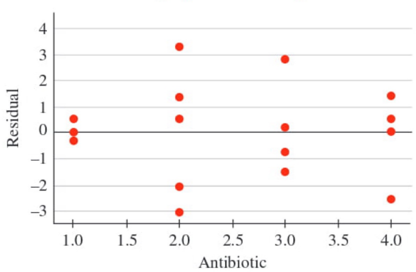
a. What design is used in collecting these data?

b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?

Normal Probability Plot
(response is Score)



Residuals Versus Antibiotic
(response is Score)



c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

a. Randomized Complete Block Design
Treatment : 4 Antibiotics
Block : child

b. (1) Normality → 无明显弯曲 or 偏离
→ 基本满足

(2) constant Variance → 参差不齐
→ 无满足
→ (c)小题中使用非参数检验更合适

c. H_0: 4种 Antibiotics 口味评分概率分布相同
H_a: 至少1种 Antibiotics 口味评分の概率与其他不同.

$$F_r = \frac{12}{b \cdot k(k+1)} \sum_{i=1}^{k} R_i^2 - 3b(k+1)$$

$b=5,\ k=4$

$\sum R_i^2 = 12^2+13^2+10^2+15^2 = 638$

$$F_r = \frac{12}{5\times4\times5} \times 638 - 3\times5\times5$$

$= 1.56$

$\chi^2_{3,0.05} = 7.815$

$\because F_r = 1.56 < 7.815$ ∉ RR

→ Fail to reject H_0.

DATA SET
DS1529

11. Yield of Wheat Exercise 9 (Chapter 11 Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

	Location						
Variety	1	2	3	4	5	6	R_i
A	35.3 4	31.0 3	32.7 2	36.8 2	37.2 3	33.1 4	18
B	30.7 1	32.2 4	31.4 1	31.7 3	35.0 2	32.7 3	14
C	38.2 5	33.4 5	33.6 3	37.1 4	37.3 4	38.2 5	26
D	34.9 3	36.1 2	35.2 5	38.3 5	40.2 5	36.0 2	22
E	32.4 2	28.9 1	29.2 4	30.7 1	33.9 1	32.1 1	10

a. Use the Friedman F_r-test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.

b. When the analysis of variance was performed in Chapter 11, the ANOVA F-test produced the test statistic $F = 18.61$ with p-value $= .000$. How do these results compare with results of the Friedman F_r-test in part a? Explain.

(a) H_0: 5种小麦品种の产量概率分布相同

H_a: 至少一种小麦品种の产量概率与其他品种不同.

$\frac{5\times(5+1)}{2} = 15$, Total $15\times 6 = 90$; $18+14+26+22+10=90$

$$F_r = \frac{12}{b\cdot k(k+1)} \sum_{i=1}^{k} R_i^2 - 3b(k+1)$$

$$= \frac{12}{6\times 5(5+1)} \times 1780 - 3\times 6\times(5+1)$$

$$= \frac{12}{180} \times 1780 - 18\times 6$$

$$= 10.67$$

$\chi^2_{0.05,4} = 9.488$

$\because 10.67 > 9.488$, $\in RR$

$\longrightarrow$ Reject H_0

(b) ANOVA $F = 18.61$, p-value $= 0.000 < 0.001$ $\longrightarrow$ significant

Friedman $F_r = 10.67$, p-value < 0.05 $\longrightarrow$ significant

$\therefore$ 2者结论完全相同.